

CoreOps Week 2 Tue Module

Kubernetes Internals & High Availability Architecture

Objective: Master control plane internals, node lifecycle, upgrade patterns, HA design, static pod behaviour, and failure-handling logic.

Tuesday: etcd HA, disaster recovery, quorum, snapshots

Critical Concepts

- etcd architecture: leader, followers, Raft log
- Why quorum matters
- etcd write path
- Snapshot vs disaster recovery
- etcdctl must-know commands
- Backup–restore strategies in production



Key Outcomes

1. Ability to restore etcd in a real cluster
2. Diagnose leader-election and latency issues
3. Know exactly when a multi-master cluster will collapse

WHAT TO PRACTICE (HANDS-ON)

All exercises run on your Kubeadm cluster (you will create this cluster in the assignment).

Lab 1: Take a snapshot

```
ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379 \
```

```
--cacert=/etc/kubernetes/pki/etcd/ca.crt \
```

```
--cert=/etc/kubernetes/pki/etcd/server.crt \
```

```
--key=/etc/kubernetes/pki/etcd/server.key \
```

```
snapshot save snapshot.db
```

Lab 2: Restore from snapshot

Stop etcd static pod, clear datadir, restore snapshot

Lab 3: Quorum failure simulation

Rename one etcd member's manifest → lose quorum → observe cluster freeze.

WHERE TO STUDY

- etcd Docs: <https://etcd.io/docs/>
 - Raft Protocol Paper (Mandatory for senior engineers)
 - Kelsey Hightower - etcd deep dive YouTube
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INFRATHRONE MINDSET NOTES

- If you break etcd, you break Kubernetes.
 - 99% of enterprise outages come from quorum loss or slow leaders.
 - Always test failure scenarios, never assume your etcd is healthy.
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